Building and publishing a native, cross-platform mobile app

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A little about me

- Full-stack developer for 8 years
- App Store published
- Outdoor lover
- Father and husband

Where we're going

- The state of mobile development
- React Native fundamentals
- Building with Expo
- Developing and testing
- Production and publishing
- A word on burnout

Idea > Technology

What’s your idea?

The state of mobile app dev

17b

Devices

257b

Downloads

Per year

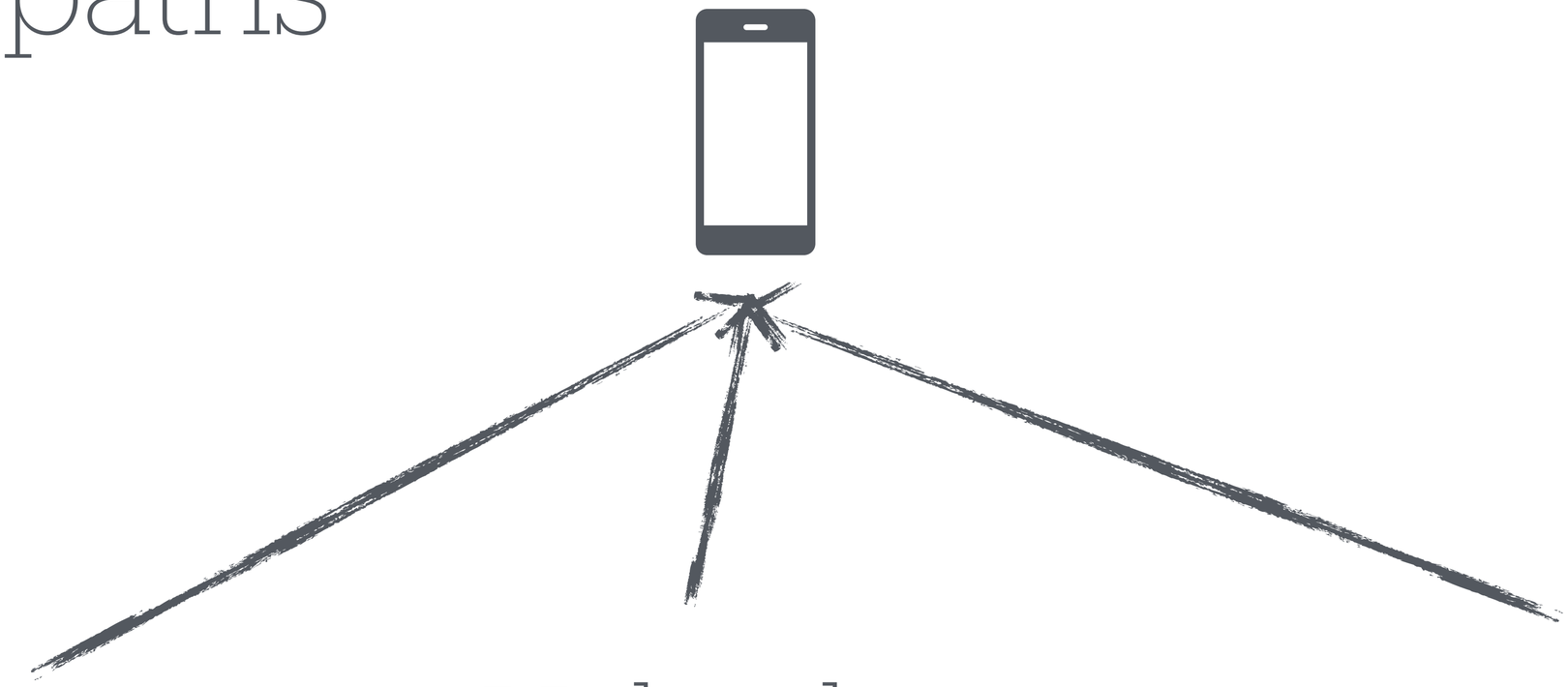
$437b

Revenue

Source: Statista as of 2023

How will you stand out?

Three paths



Native

- Best performance
- Double the effort
- Native UX

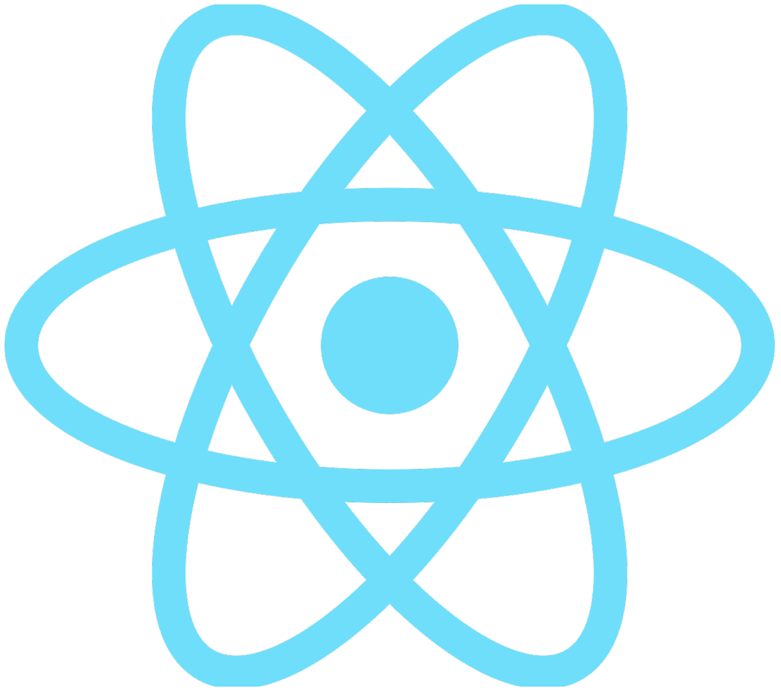
Hybrid

- Slow performance
- Least effort
- Generic UX

Cross-platform

- Near-native performance
- Efficient development
- Native UX

The landscape of cross-platform



React Native - Facebook

Flutter - Google

React Native Fundamentals

What you get

- Routing
- Basic components
- Common languages and design patterns
- Vibrant community support
- Approachable learning curve from web

Web vs Native

Web

```
<div className="container">
  <h1>Bingo is fun!</h1>
  <button onClick={handleClick}>
    Play now
  </button>
</div>
```

Native

```
<View style={styles.container}>
  <Text>Bingo is fun!</Text>
  <TouchableOpacity onPress={handlePress}>
    <Text>Play now</Text>
  </TouchableOpacity>
</View>
```

Web vs Native

Web

```
.container {
  display: flex;
  justify-content: center;
  margin: 20px;
}
```

Native

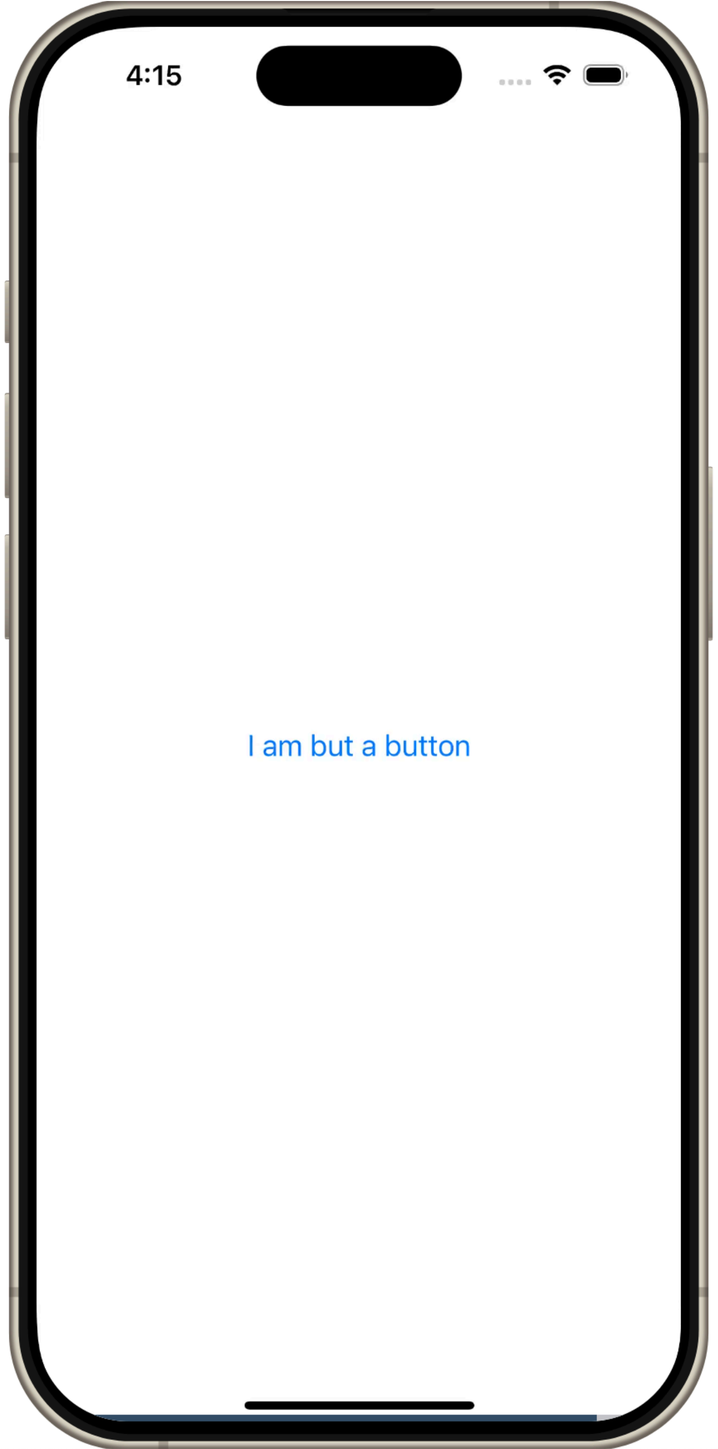
```
const styles = StyleSheet.create({
  container: {
    flexDirection: 'column',
    justifyContent: 'center',
    margin: 20,
  }
});
```

Common Components

```
<View>
 <Text style={styles.bodyText} >
  I am but a small text
  {'\n'}
  {'\n'}
 </Text>
</View>
```

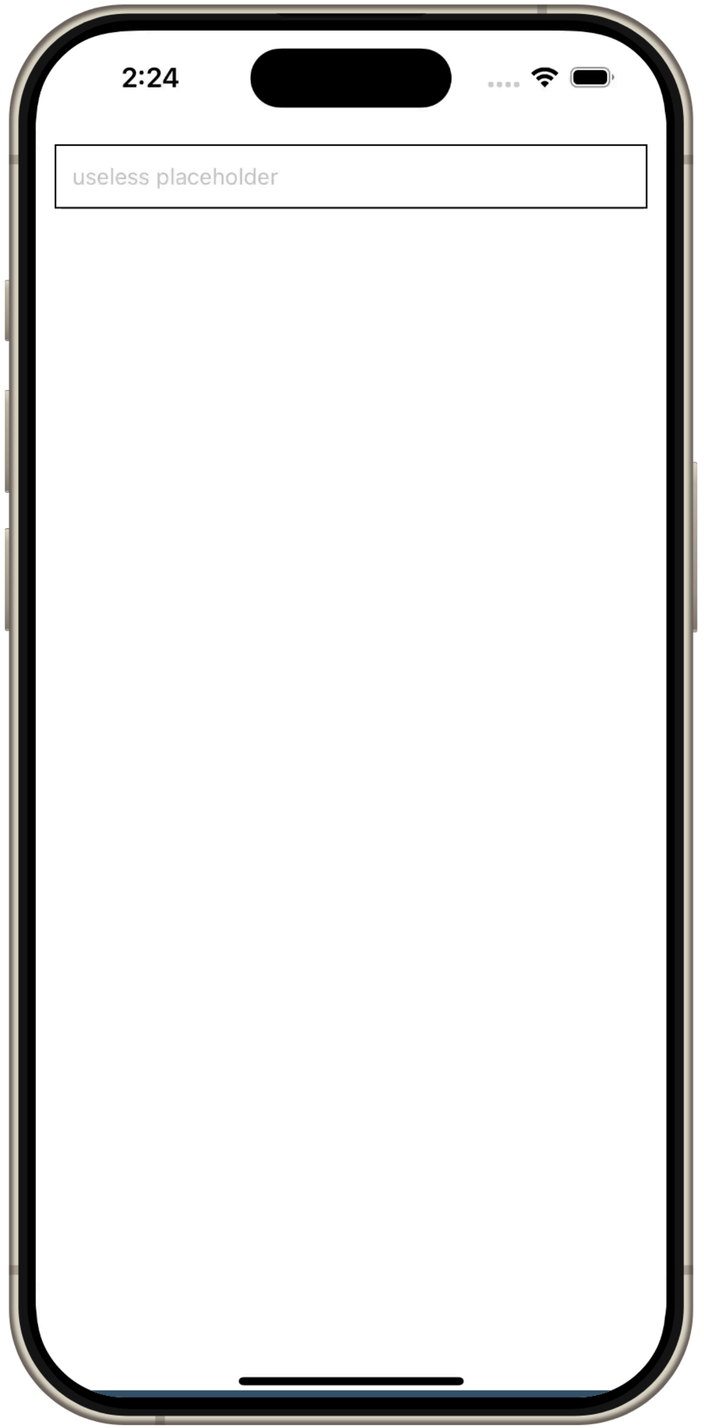
Common Components

```
<View style={styles.outline}>
  <Button
    title="I am but a button"
    onPress={() => Alert.alert('You won!')}
  />
</View>
```



Common Components

```
<SafeAreaView>
 <TextInput
  style={styles.input}
  onChangeText={onChangeNumber}
  value={number}
  placeholder="useless placeholder"
  keyboardType="numeric"
 />
</SafeAreaView>
```



Building with Expo

React Native

Simplifies the coding

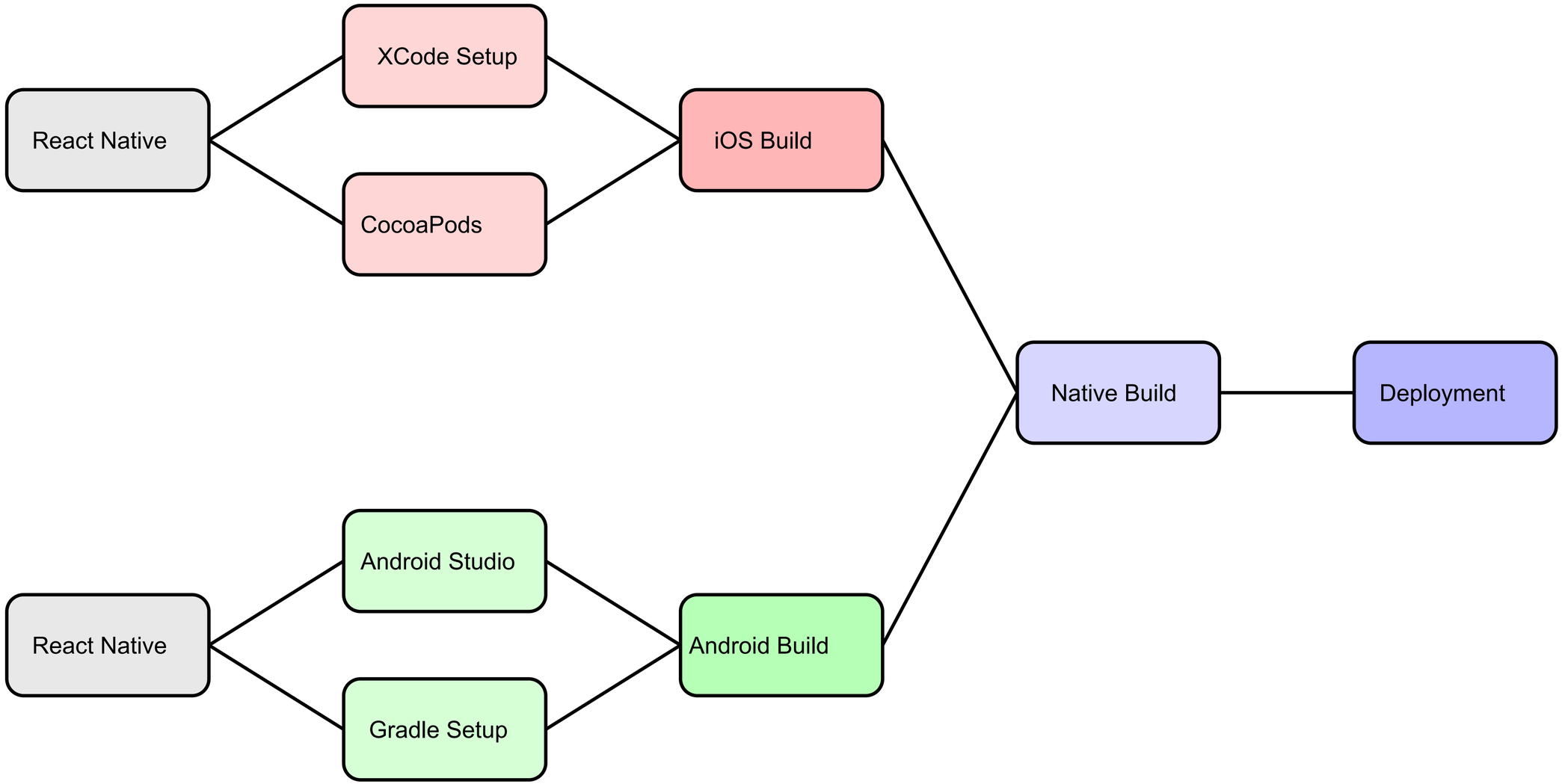
Expo

Simplifies the processes

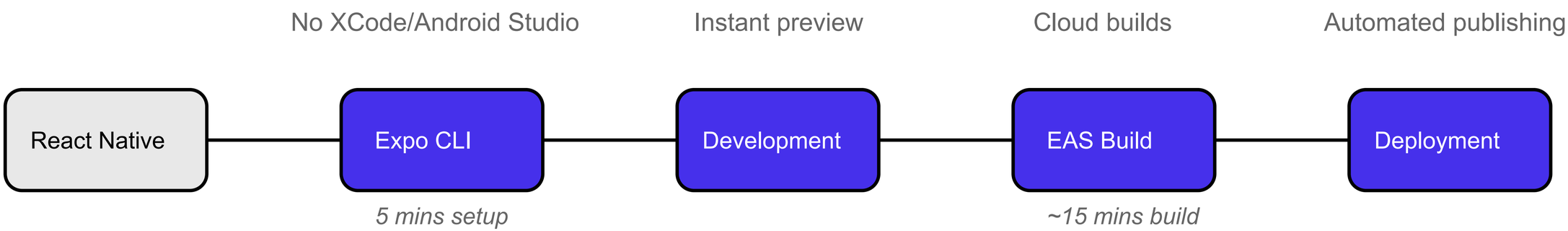
Without Expo

- Complex local setup
- Complex builds
- Tedious and lengthy submission

Without Expo

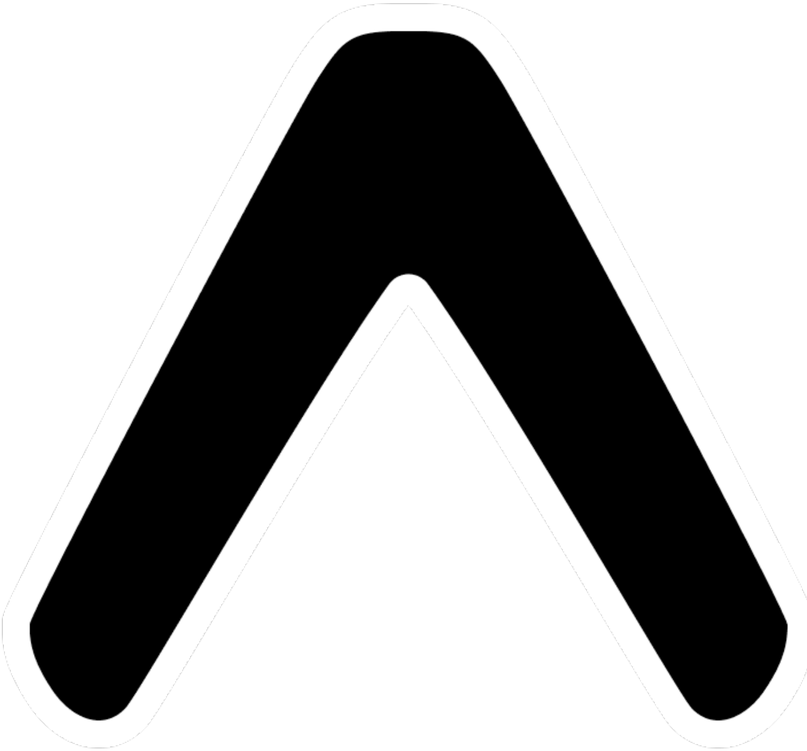


With Expo


No XCode/Android Studio
Instant preview
Cloud builds
Automated publishing
React Native
Expo CLI
Development
EAS Build
Deployment
5 mins setup
~15 mins build

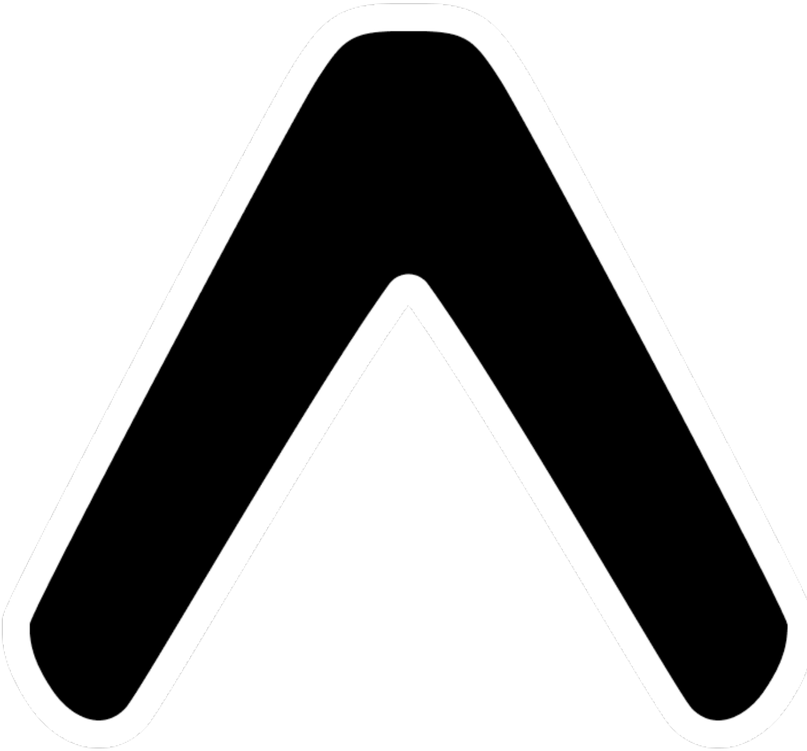
What Expo offers

- Streamlined local setup
- Automated builds
- Simplified local and device testing
- Guided, automated app submission
- Free* and open source



Additional features in Expo

- Comprehensive CLI
- Fast refresh
- Native API access
- End to end testing



Native API Access

Via Expo

```
import * as Haptics from 'expo-haptics';

function triggerBingo() {
  playVictorySound();
  Haptics.impactAsync(Haptics.ImpactFeedbackStyle.Medium);
  setIsBingo(true);
  removeData(board.id.toString());
}
```

Native API Access

Via Expo

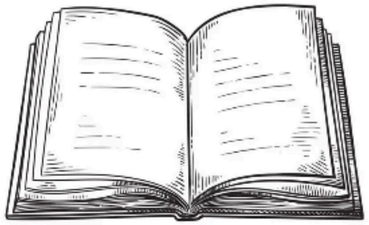
```
import { Audio } from 'expo-av';

async function playVictorySound() {
  if (isSoundEnabled && !isGameOver) {
    const { sound } =
      await Audio.Sound.createAsync( require('../assets/victory.mp3')
    );
    await sound.playAsync();
  }
}
```

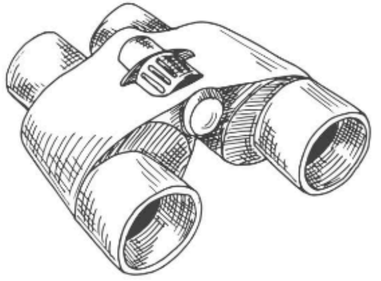
From idea to development

Idea

Storyboard



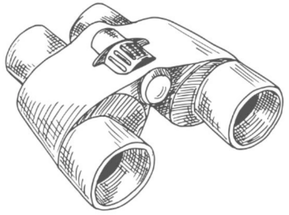
Plan ahead



Develop

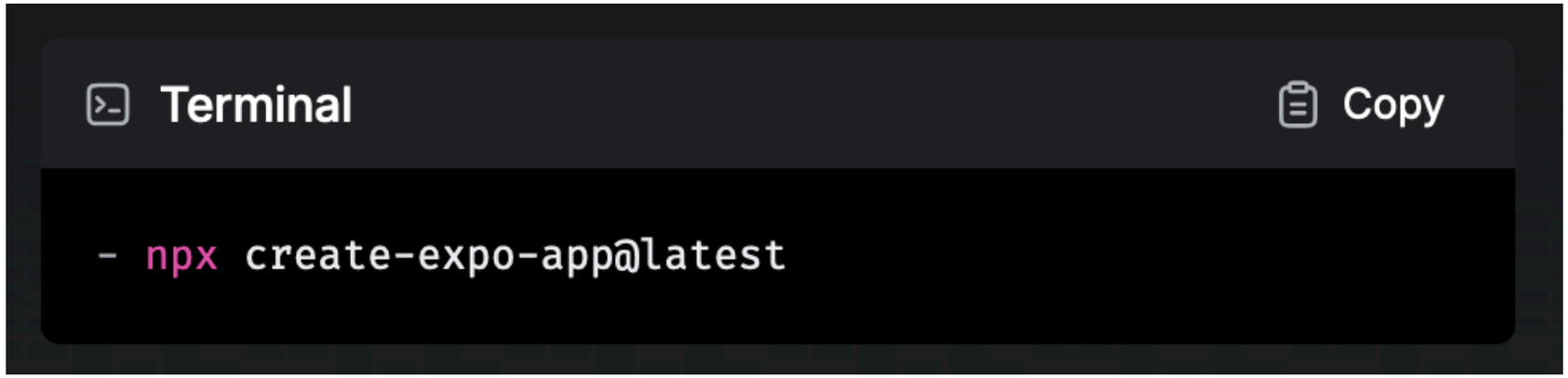
Publish

Plan ahead



ERS: LAST 7 DAYS USING MEDIAN
AD TIME VS BOUNCE RATE
OPTIONS
Median Page Load (LUX): 2.056s
Bounce Rate
7s
57.1%
100 %
80 %
60 %
40 %
20 %
0 %
0
2.5
5
7.5
10
12.5
15
17.5
Page Load (LUX)
Bounce Rate
START RENDER VS BOUNCE RATE
OPTIONS
40K
32K
24K
16K
8K
0
Median Start Render (LUX): 1.031s
0
1
2
3
4
5
Start Render (LUX)
Bounce Rate
GE VIEWS VS ONLOAD
OPTIONS
ge Load (LUX)
.7s
Page Views (LUX)
2.7Mpvs
Bounce Rate (LUX)
40.6%
500K 100%
400K 80%
300K 60%
200K 40%
SESSIONS
OPTIONS
Sessions (LUX)
479K
Session Length (LUX)
17min
PVs Per Session (LUX)
2pvs
4 pvs
3.2 pvs
2.4 pvs
1.6 pvs
100K
80K
60K
40K

Develop

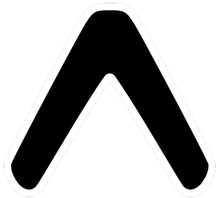


docs.expo.dev/get-started

Publishing

Prerequisites

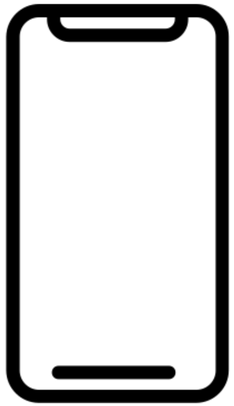
Developer Accounts



Account	$	Term
Apple Developer	99	Yearly
Google Play Console	25	Lifetime
Google Cloud Service Account	Free	Lifetime
Expo.dev	Free	Lifetime

Get your docs in a row

privacypolicies.com



Screenshots

Iconography

Writeup

Prerequisites

EAS CLI

```
npm install -g eas-cli && eas login
```

docs.expo.dev/submit/introduction/

Android

Google Play

API account

- Google Service account
  - Needed for automated submission
  - Create and download JSON key
    - Take note of file location

Google Play

App Store submission

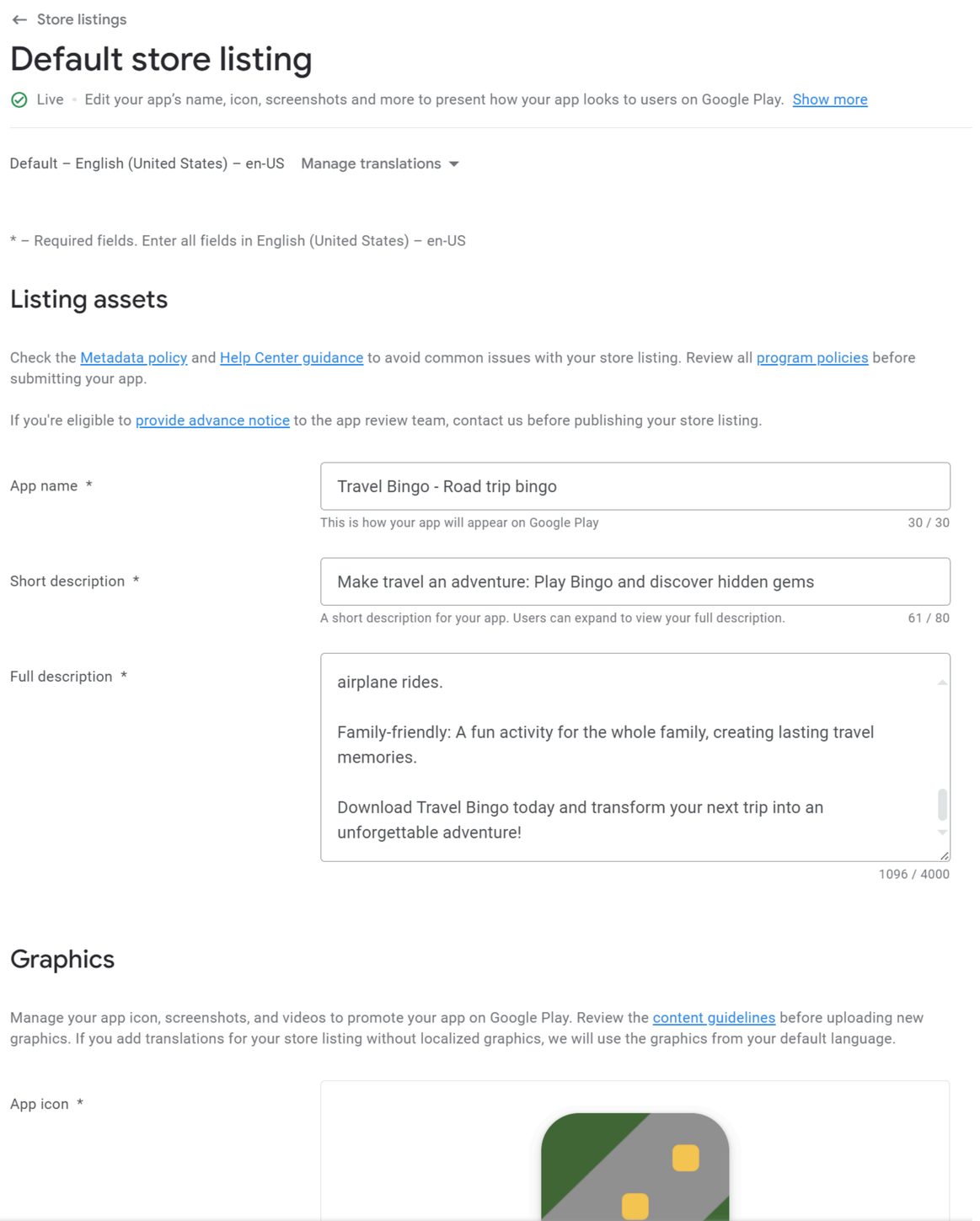
app.json

```
"android": {
  "package": "com.carrollmedia.travelbingo",
  "versionCode": 7
}
```

Google Play Console

App Store submission

- Category
- Age rating survey
- Privacy declaration



Google Play

App Store submission

eas build --platform android

apk

~~eas submit~~

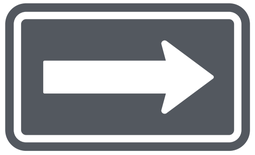
Google Play Console

Google Play

Closed Beta

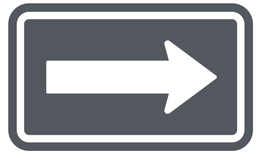
12

Users



14

Days



Production app review

Pre survey

Post survey

Apple

Apple App Store

Privacy Declarations

```
    "ios": {
      "supportsTablet": true,
      "bundleIdentifier": "com.carrollmedia.travelbingo",
      "infoPlist": {
        "NSPrivacyAccessedAPITypes": [
          {
            "NSPrivacyAccessedAPIType": "NSPrivacyAccessedAPICategorySystemBootTime",
            "NSPrivacyAccessedAPITypeReasons": [
              "8FFB.1"
            ]
          },
        ]
      }
    },
```

app.json

Apple App Store

Who needs privacy?

Saving data?

↓

`…DiskSpace`

Using a timer?

↓

`…SystemBootTime`

Apple App Store

Submission

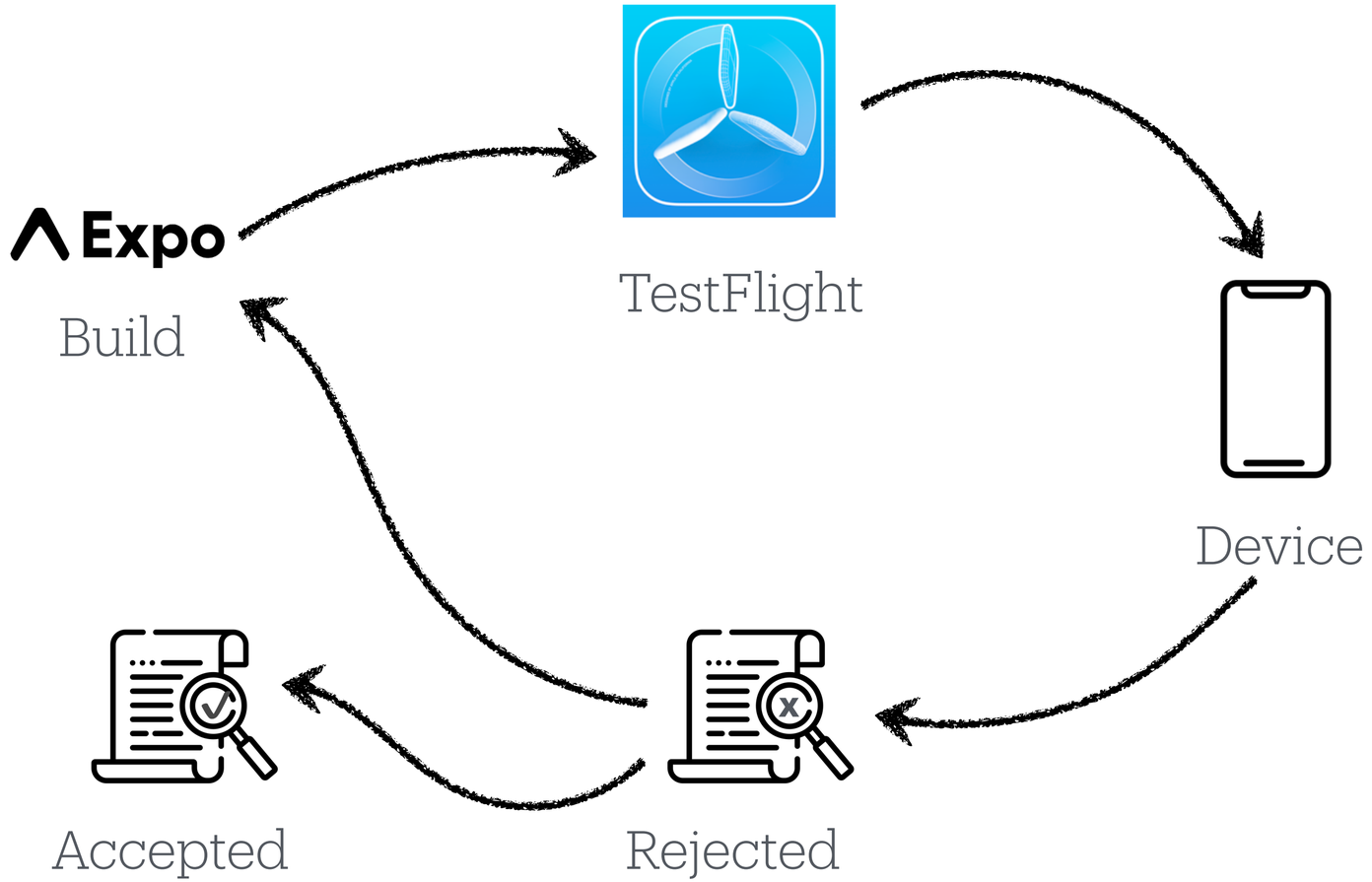
App Store Connect

eas build --platform ios

eas submit

Apple App Store

Testing flow

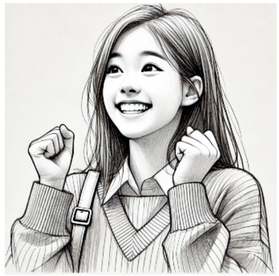


Your app is live!

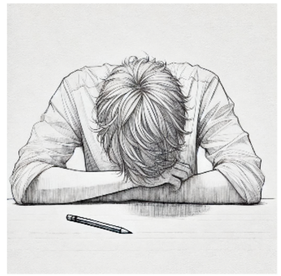
Let’s talk about burnout

The project death cycle

Idea



Excitement



Slog

Distraction

Shelf

Tips for success

- Balance your time
- Be patient
- Plan to get stuck
- Don’t compare yourself to others
- Involve others

“The journey of a thousand miles begins with a single step”

- Lao Tzu

Questions?

carrollmedia.dev/react-native

Appendix

Devices - https://www.statista.com/statistics/245501/multiple-mobile-device-ownership-worldwide/
Downloads - https://www.statista.com/statistics/271644/worldwide-free-and-paid-mobile-app-store-downloads/
Revenue - https://www.statista.com/outlook/amo/app/worldwide

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